

# The algebraic monodromy of Lagrangian fibrations

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# Holomorphic symplectic manifolds

## Definition

A *symplectic manifold*  $(M, \omega)$  is a smooth manifold  $M$  equipped with a closed non-degenerate global 2-form  $\omega \in H^0\left(M, \bigwedge^2 T^*M\right)$ .

Contracting with  $\omega$  induces an isomorphism  $\iota_\omega : TM \xrightarrow{\sim} T^*M$ .

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Note that in the latter two examples, despite  $X$  being a complex manifold, the Kähler form  $\omega \sim dz_1 \wedge d\bar{z}_1 + \cdots + dz_n \wedge d\bar{z}_n$  is not holomorphic.

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- Any product of the spaces above is symplectic.

# The decomposition theorem

We can generalize the K3 surface as follows:

## Definition

We say a compact holomorphic symplectic manifold  $(X, \sigma)$  is *irreducible* if it is simply connected and  $\bigoplus_{p=0}^{\dim X} H^0(X, \Omega_X^{2p}) = \mathbb{C}[\sigma]$ .

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## Beauville–Bogomolov Decomposition Theorem

Let  $(X, \sigma)$  be a compact holomorphic symplectic manifold. Then there exists a finite cover

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Since complex tori are somewhat well understood, we will focus on understanding the irreducible holomorphic symplectic (IHS)  $X_i$ .

## Definition

Let  $(X, \sigma)$  be a holomorphic symplectic manifold of dimension  $2n$ , and let  $Y \subseteq X$  be a subvariety.

- 1  $Y$  is *isotropic* if  $\sigma|_Y = 0$ .
- 2  $Y$  is *Lagrangian* if  $Y$  is maximally isotropic, i.e.  $\sigma|_Y = 0$  and  $\dim Y = n$ .



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## Theorem (Greb–Lehn '14)

*Let  $f : X \rightarrow B$  be a proper surjective morphism,  $B$  a normal analytic space of dimension not equal to 0 or  $2n$ . Then the fibers of  $f$  are Lagrangian subvarieties and  $B$  is a projective variety of dimension  $n$ .*

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For this reason, we refer to any such morphism  $f : X \rightarrow B$  as a *Lagrangian fibration*. Understanding the geometry of Lagrangian fibrations will help us understand the geometry of IHS manifolds.

# The structure of the fibration

Let  $f^\circ : X^\circ \rightarrow B^\circ$  be the smooth fibers of  $f : X \rightarrow B$ . We need to understand

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By Hodge theory, an Abelian variety  $A$  can be recovered from its cohomology:

$$A \simeq H^{1,0}(A)/(H^1(A, \mathbb{Z}))^\vee.$$

We will study the monodromy representation

$$\rho : \pi_1(B^\circ, b) \rightarrow \text{Aut}(H^1(X_b, \mathbb{Z})).$$

# Algebraic monodromy groups

## Definition

The *full algebraic monodromy group*  $M$  of  $\rho$  is the Zariski closure of  $\rho(\pi_1(B^\circ, b)) \subseteq \text{Aut}(H^1(X_b, \mathbb{Q})) = \text{Sp}_{2n, \mathbb{Q}}$ . Its identity component  $M^\circ \subseteq M$  is the *(connected) algebraic monodromy group*.



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Now we can classify the possible representations of the algebraic monodromy group.

## Theorem (V '26+)

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- 3  $M^\circ = \{1\}$ , acting trivially on  $H^1(X_b, \mathbb{Q})$ .

# Geometric consequences

In each of the cases above, we extract information regarding the geometry of  $X$  and  $f : X \rightarrow B$ .

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The case (3) is also somewhat well understood based on previous work:

## Theorem (Kim–Laza–Martin '25, V '26)

*When  $M^\circ = \{1\}$ , there exists an elliptic curve  $E$  for which the smooth fibers  $X_b$  are isogenous to  $E^n$ .*



# Proof of the main theorem.

*Proof idea.* We have the following isomorphism of bundles:

$$\begin{array}{ccccccc} 0 & \longrightarrow & \mathcal{T}_{X^\circ/B^\circ} & \longrightarrow & \mathcal{T}_{X^\circ} & \longrightarrow & f^*\mathcal{T}_{B^\circ} \longrightarrow 0 \\ & & \parallel & & \parallel & & \parallel \\ 0 & \longrightarrow & f^*\Omega_{B^\circ}^1 & \longrightarrow & \Omega_{X^\circ}^1 & \longrightarrow & \Omega_{X^\circ/B^\circ}^1 \longrightarrow 0 \end{array}$$

The fibers of  $f$  are given by quotienting  $f_*\Omega_{X^\circ/B^\circ,b}^1$  by a full rank sublattice. In particular, the structure of the fibers of  $f$  comes from  $\mathcal{T}_B$ .

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There exists a cover  $\pi : B' \rightarrow B$  for which  $\mathcal{T}_{B'} = \mathcal{F}_1 \oplus \cdots \oplus \mathcal{F}_k$  splits into transverse foliations  $\mathcal{F}_i$ , which are permuted by the covering group  $\mathfrak{S}_k$ .

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$$p : B' \rightarrow Z^k.$$

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*Proof idea (continued).* Now quotient  $p : B' \rightarrow Z^k$  by the symmetric group to get  $\bar{p} : B \rightarrow \mathrm{Sym}^k Z$ . This induces a surjection on intersection cohomology:

$$\mathrm{IH}^*(\mathrm{Sym}^k Z, \mathbb{Q}) = \mathrm{Sym}^k(\mathrm{IH}^*(Z, \mathbb{Q})) \subseteq \mathrm{IH}^*(B, \mathbb{Q}).$$

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By Matsushita,  $\text{IH}^*(B, \mathbb{Q}) = H^*(\mathbb{P}^n, \mathbb{Q})$ . Therefore

$$\text{IH}^*(Z, \mathbb{Q}) = H^*(\mathbb{P}^1, \mathbb{Q}),$$

forcing  $Z = \mathbb{P}^1$  and  $k = n$ . □

# Questions